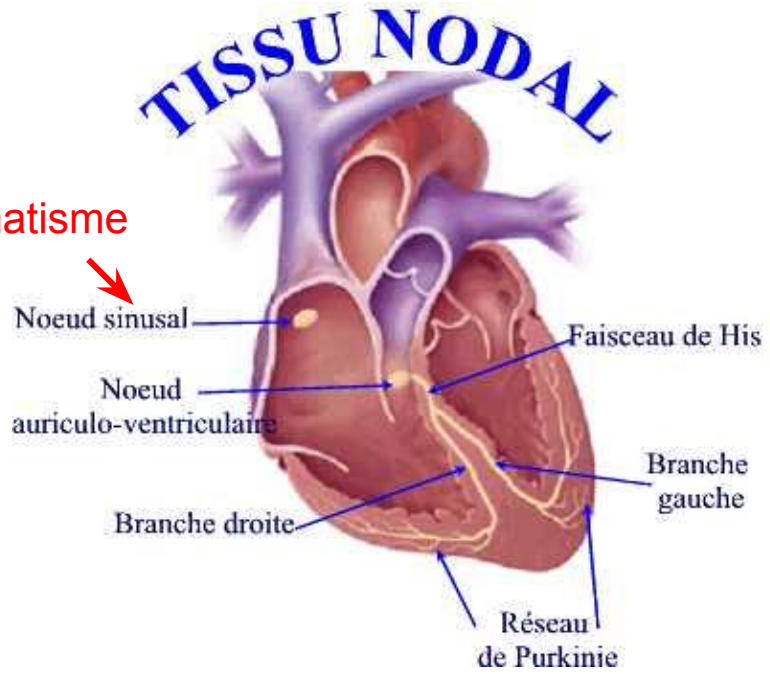
Les bases du tracé ECG

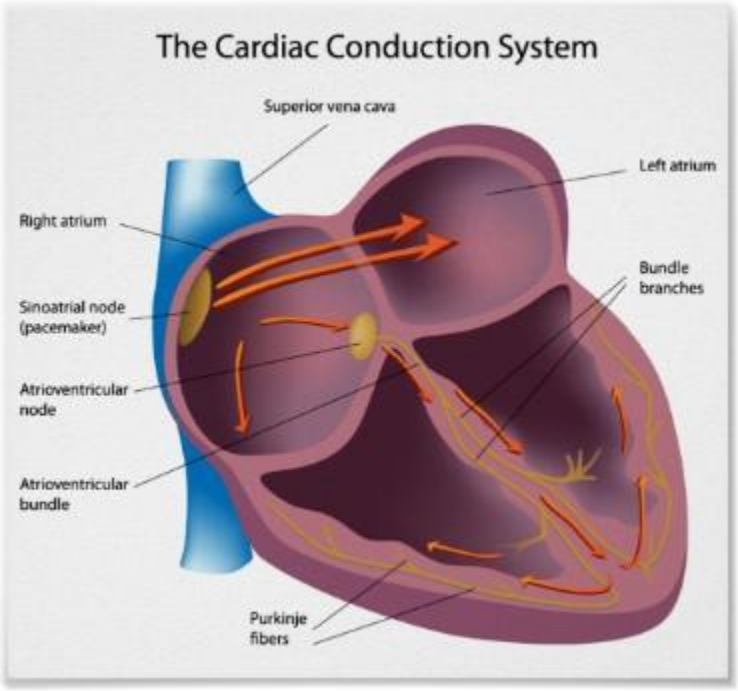
Pr. S.BENSAFIR

La conduction électrique cardiaque

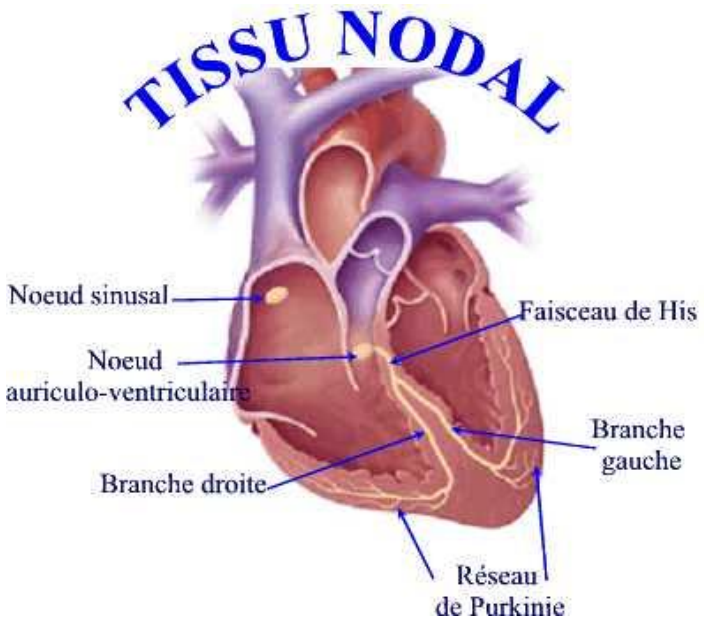
Cardiomyocytes: petites centrales électriques

Automatisme

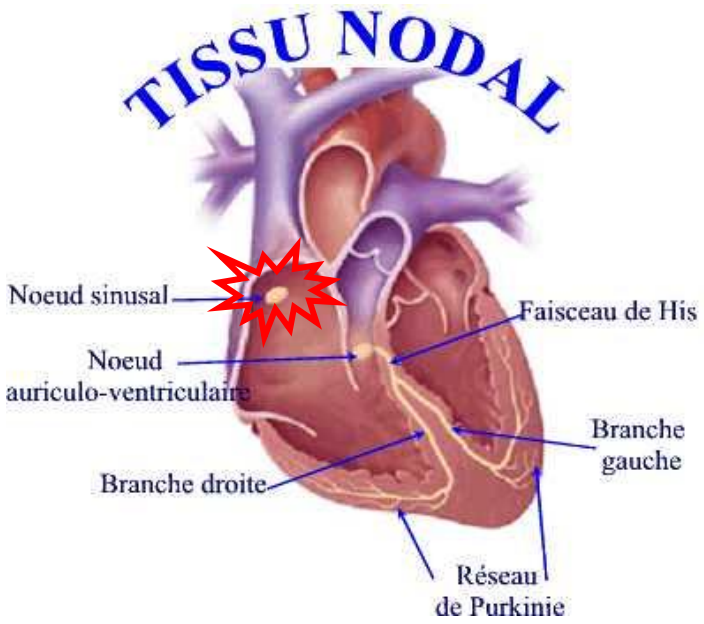


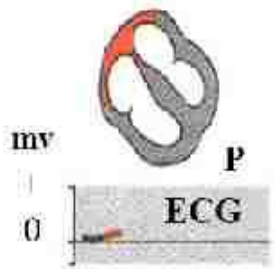


Propagation de l'onde et effets

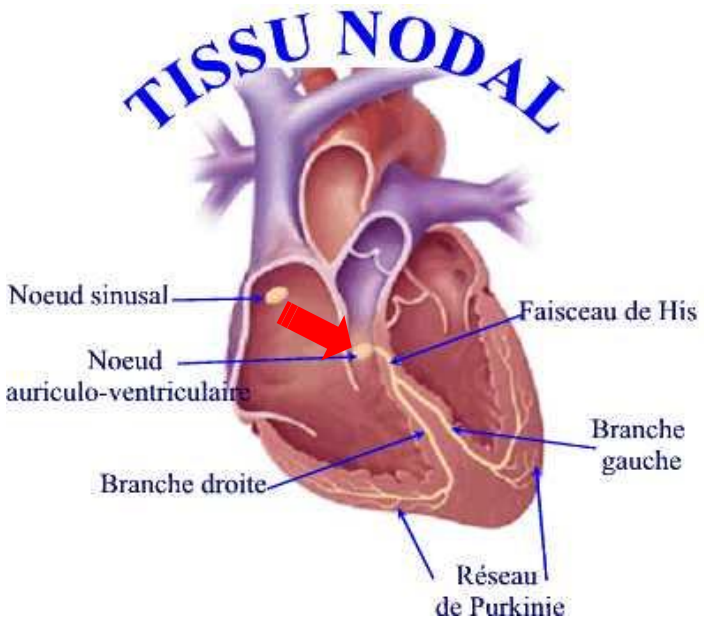


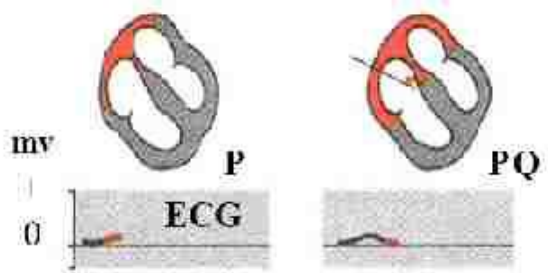
Propagation de l'onde et effets



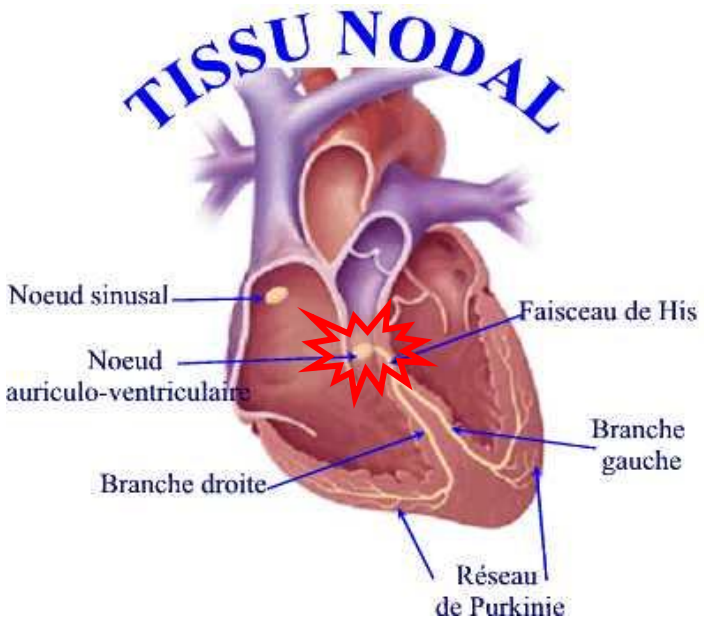


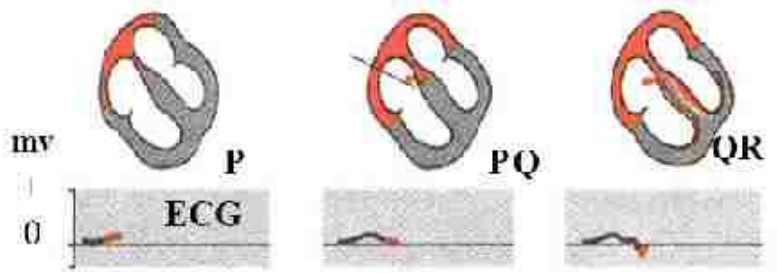
Propagation de l'onde et effets



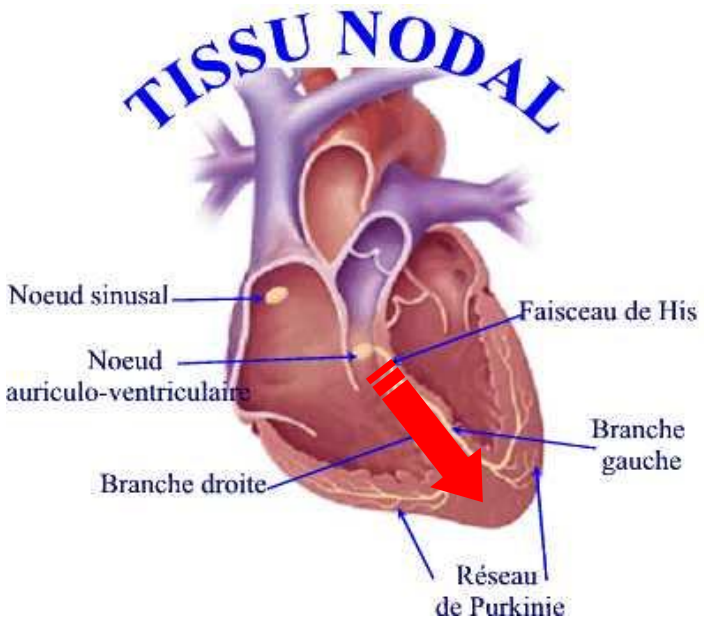


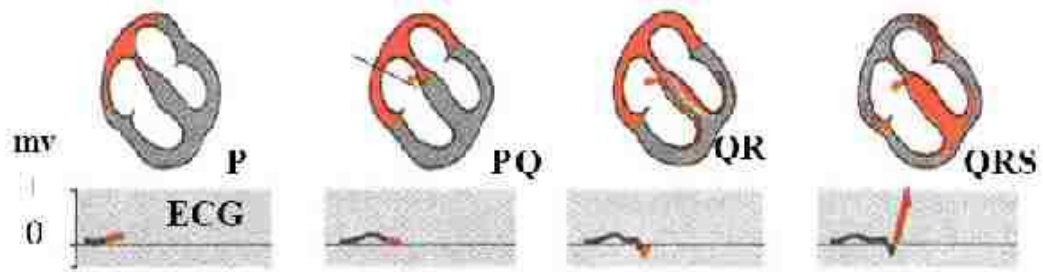
Propagation de l'onde et effets



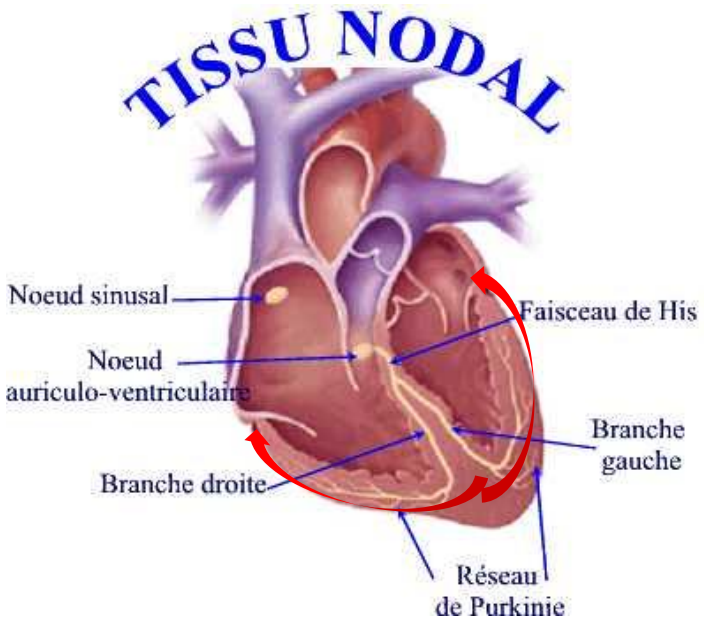


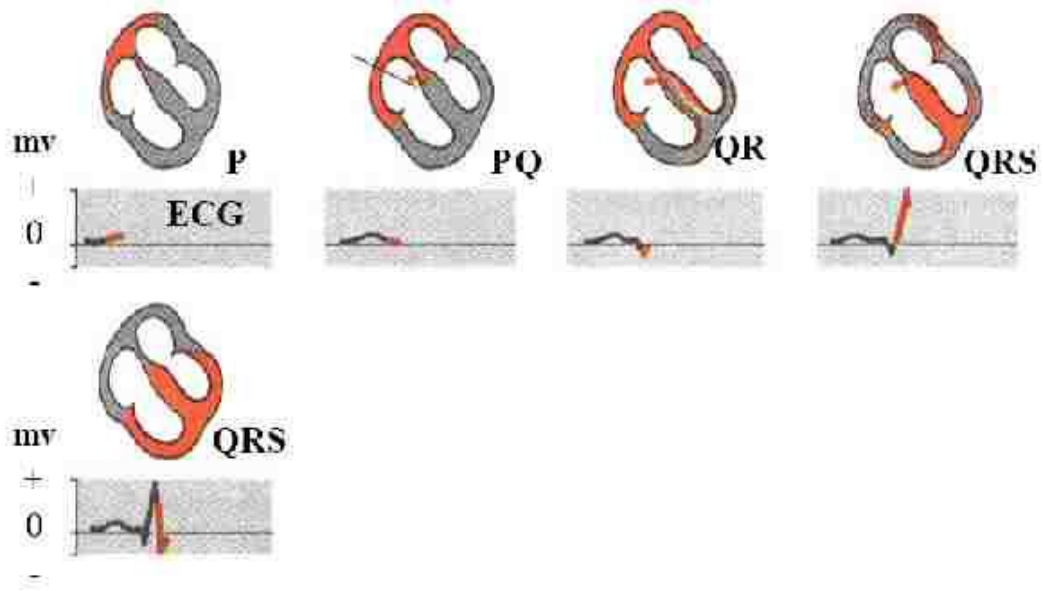
Propagation de l'onde et effets



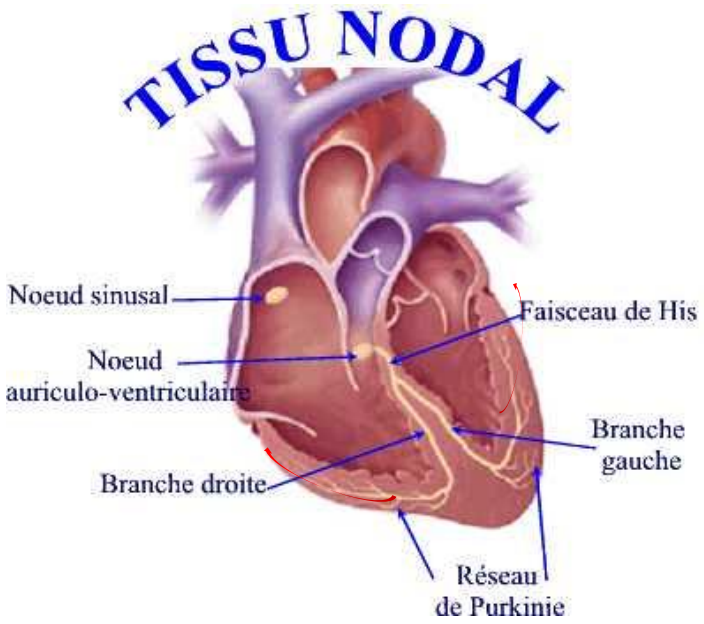


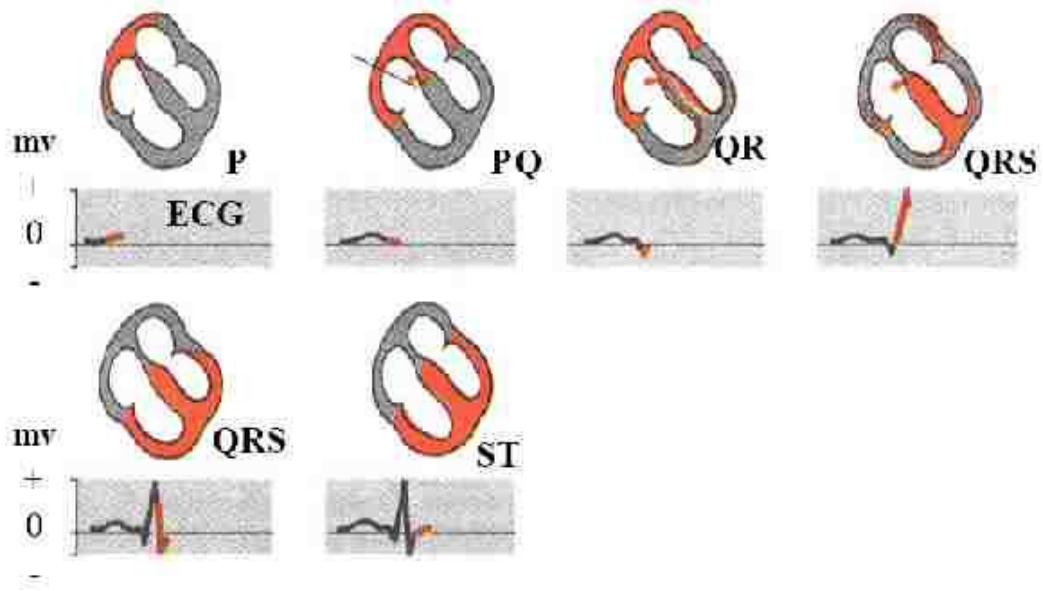
Propagation de l'onde et effets



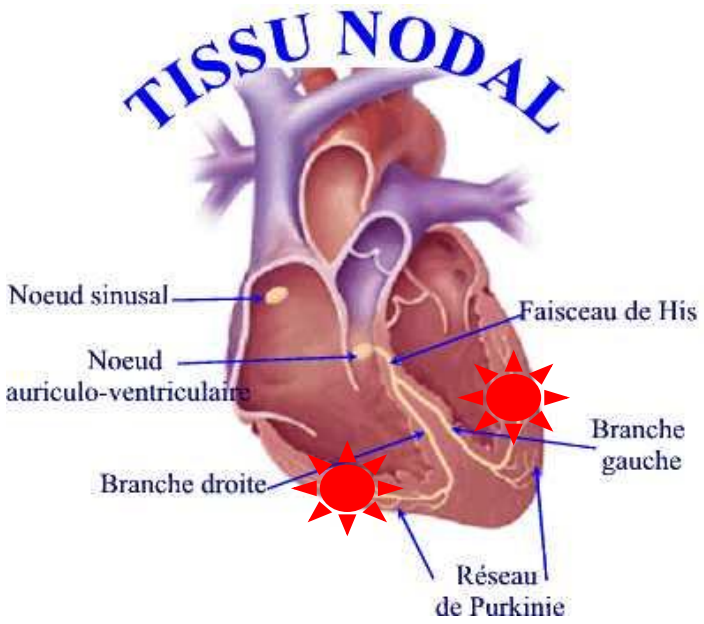


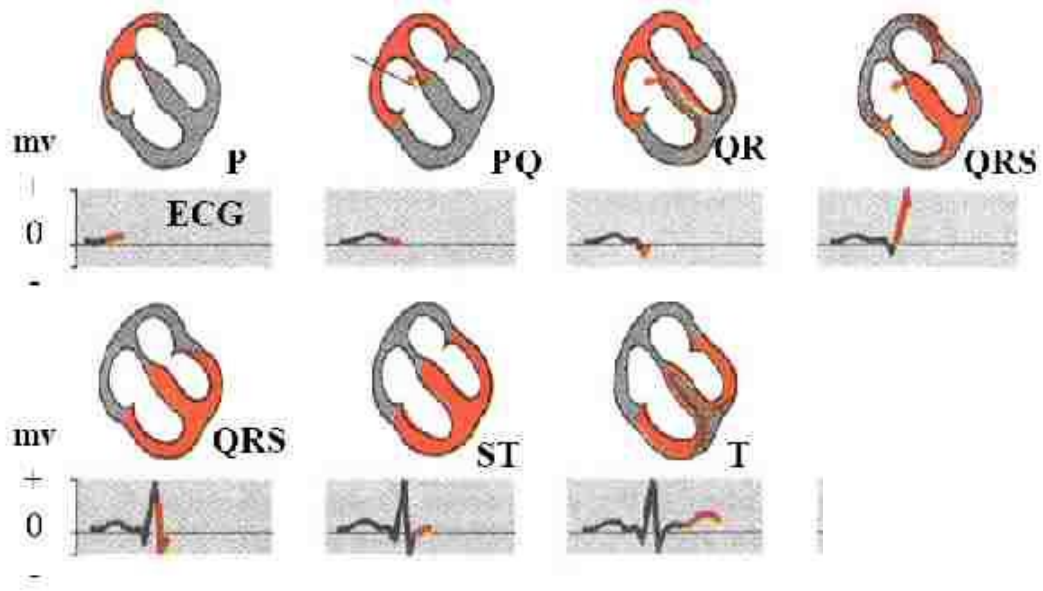
Propagation de l'onde et effets



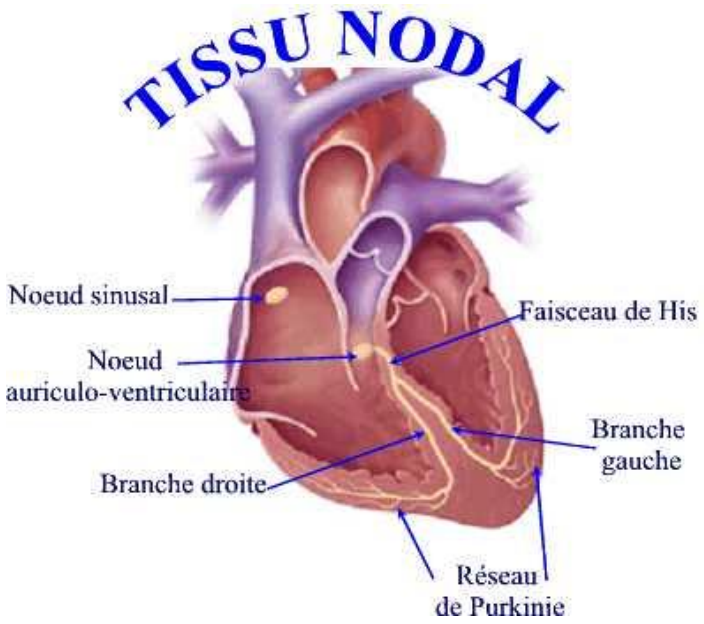


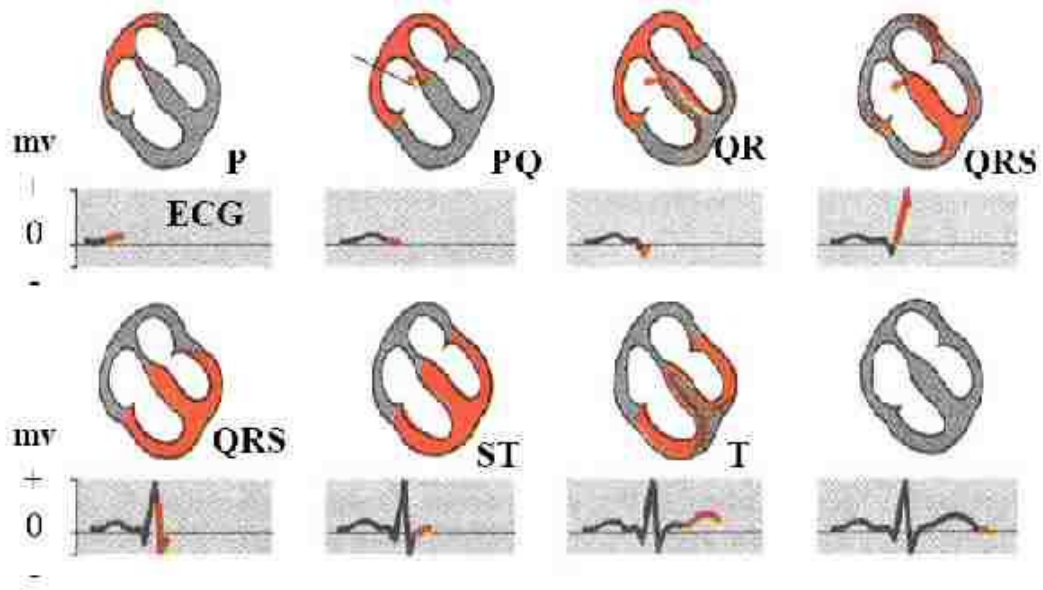
Propagation de l'onde et effets



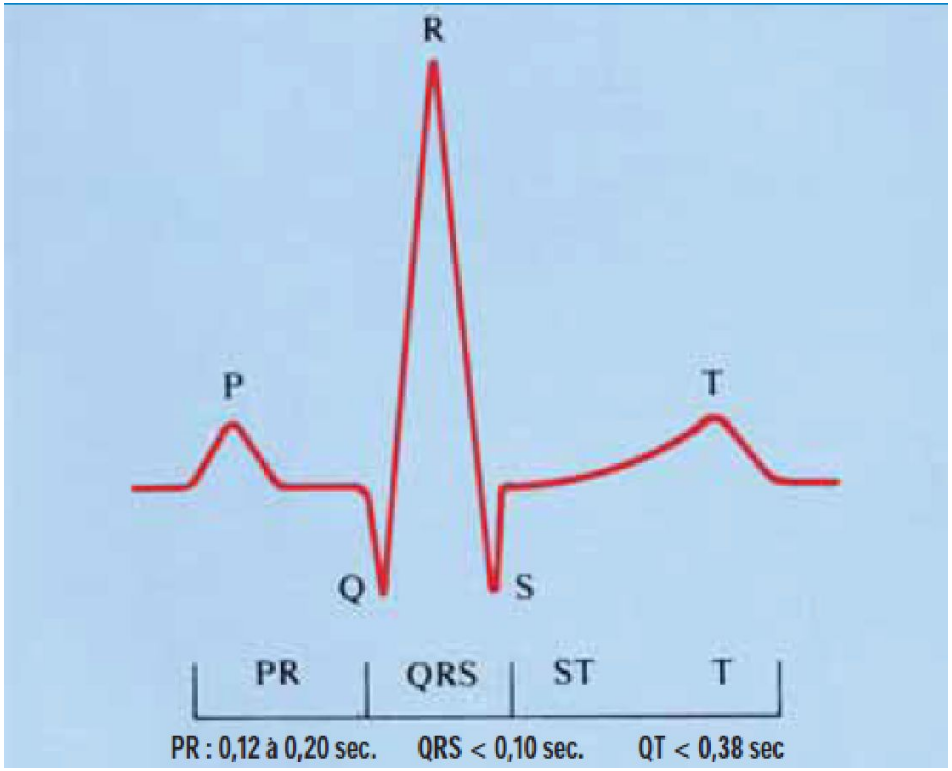


Propagation de l'onde et effets

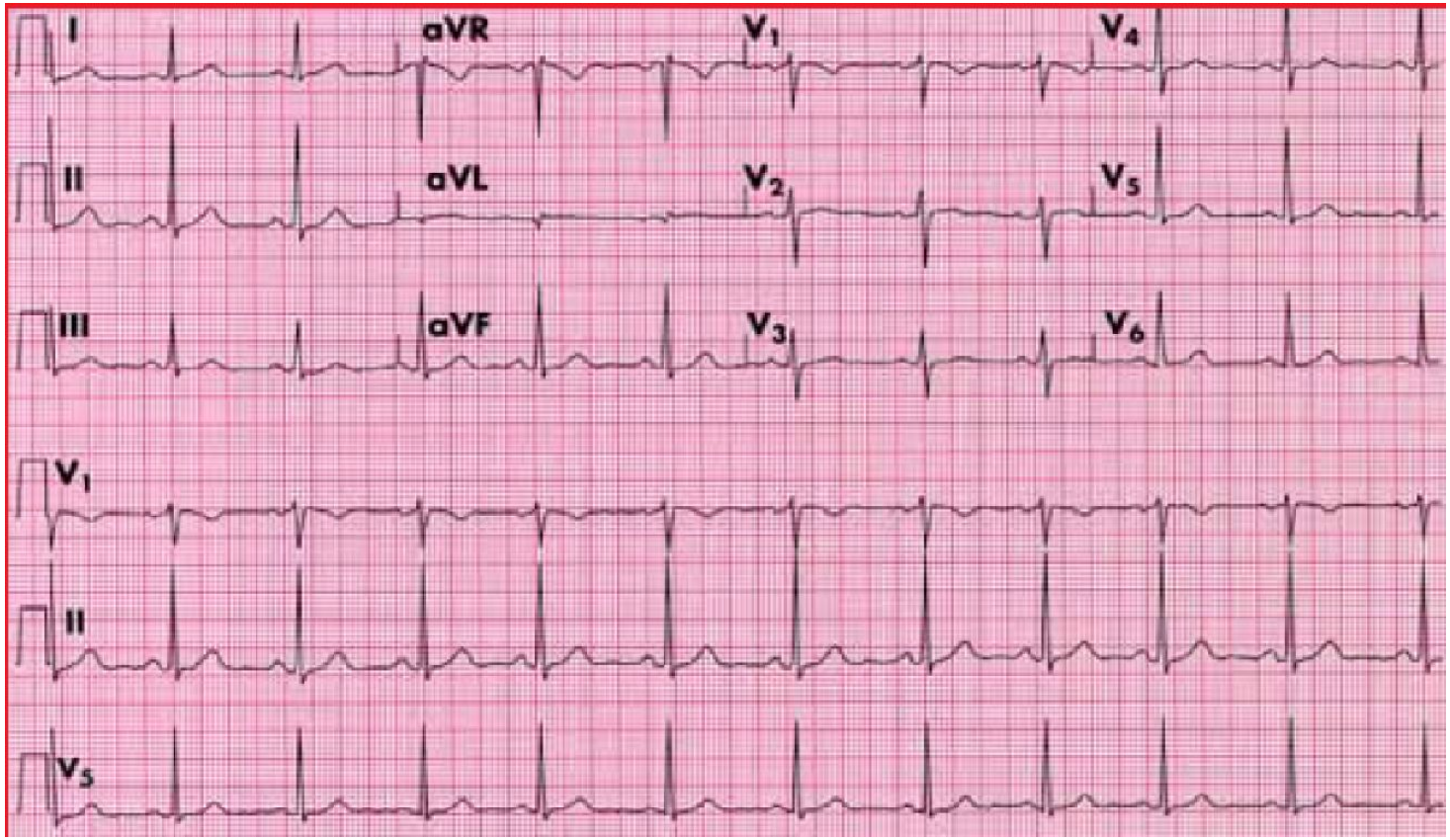




Cycle P-QRS-T



ECG normal



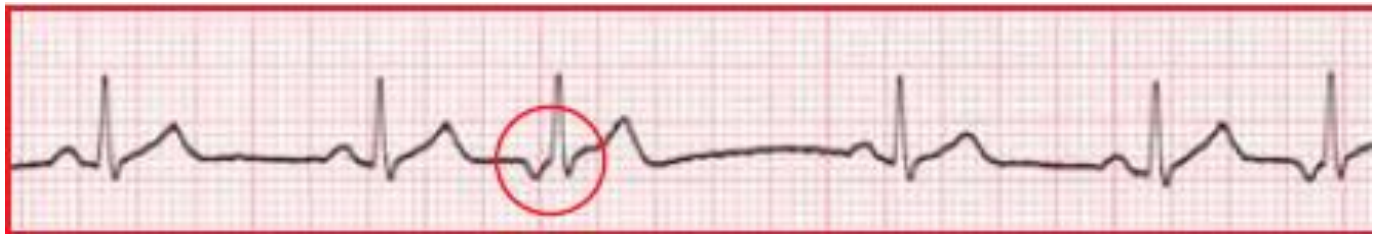
- Chaque P est suivie d'un QRS
- Chaque QRS est précédé d'une onde P
- Rythme est régulier (…)

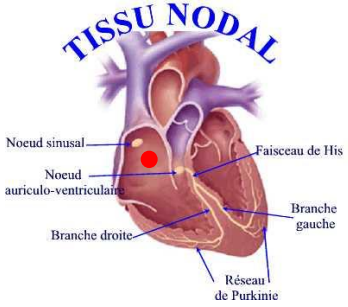
Anomalies au niveau de l'onde P

Normal









1 Onde P d'aspect différent

N'arrivant pas au bon temps → Extrasystole auriculaire

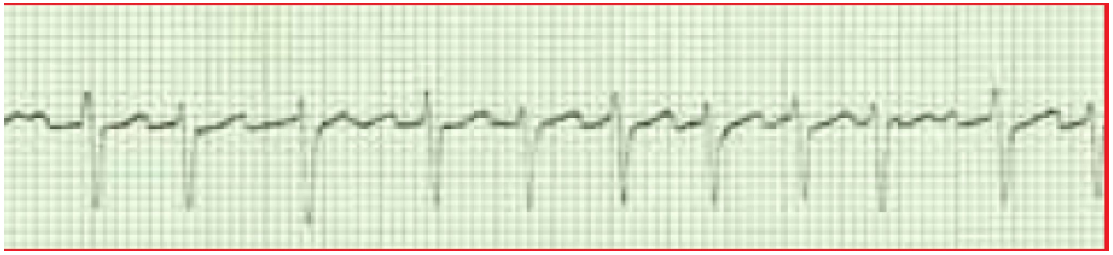
QRS normal

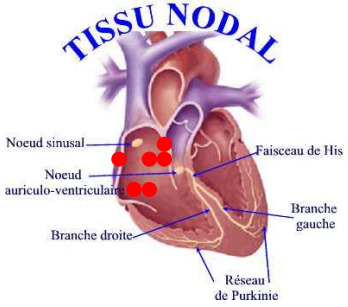
Anomalies au niveau de l'onde P

Normal









Pas d'onde P

Rythme irrégulier

Aspect QRS normal

→ Fibrillation atriale

Anomalies au niveau de l'onde P

Normal







Pas d'onde P

Rythme irrégulier → Fibrillation atriale

Aspect QRS normal

Anomalies au niveau de l'onde P

Normal



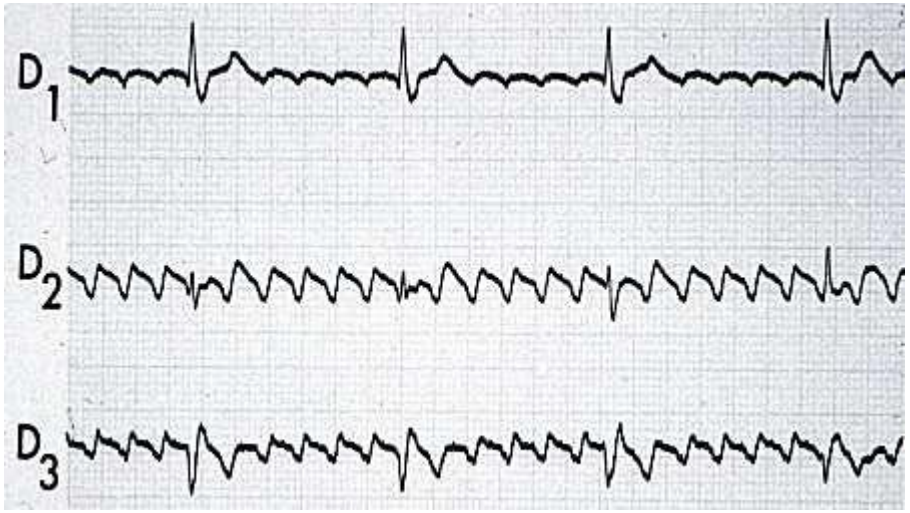


Pas d'onde P

Aspect en « toit d'usine »

Rythme régulier

Aspect QRS normal



→ Flutter auriculaire

Anomalies au niveau de l'onde P

Normal





Pas d'onde P

Aspect en « toit d'usine »

Rythme régulier

Aspect QRS normal



→ Flutter auriculaire

Anomalies au niveau de l'onde P

Normal



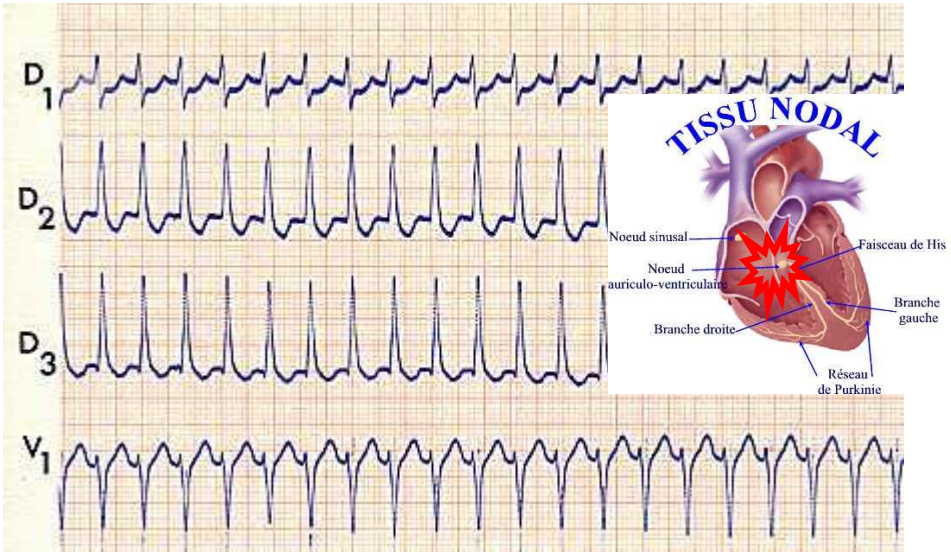


Pas d'onde P

Rythme régulier

Aspect QRS normal

Tachycardie très rapide



→ Tachycardie jonctionnelle

Anomalies au niveau du QRS

Normal

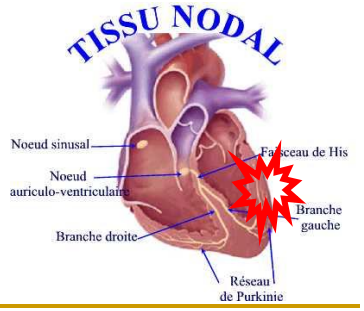




Ondes P et QRS

1 QRS différent, large

→ Extrasystole ventriculaire



Anomalies au niveau du QRS

Normal

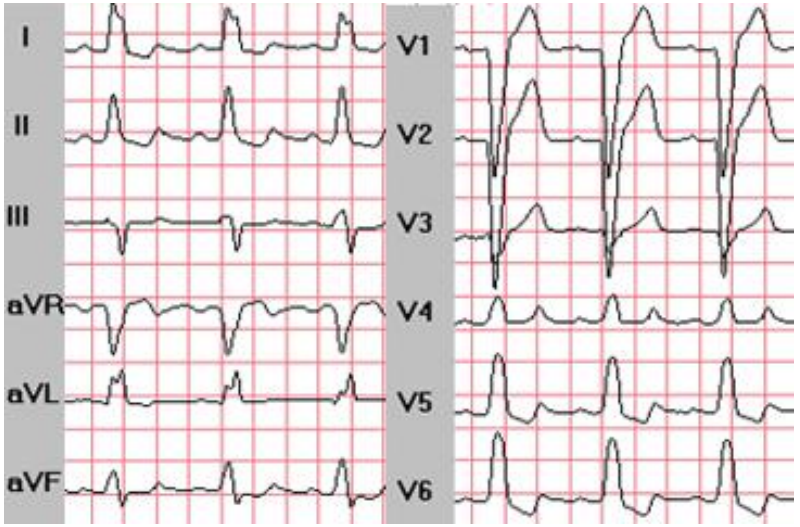


Ondes P et QRS

Mais QRS larges

Tous identiques

→ Bloc de branche



Anomalies au niveau du QRS

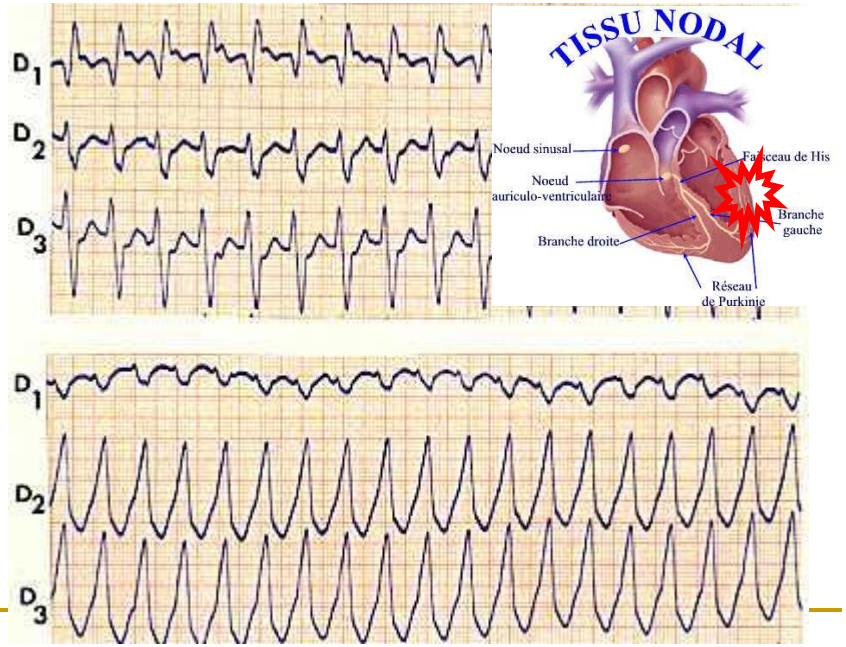
Normal



Pas d'onde P

QRS larges tous pareils

Rythme rapide régulier



→ Tachycardie ventriculaire

Anomalies au niveau du QRS

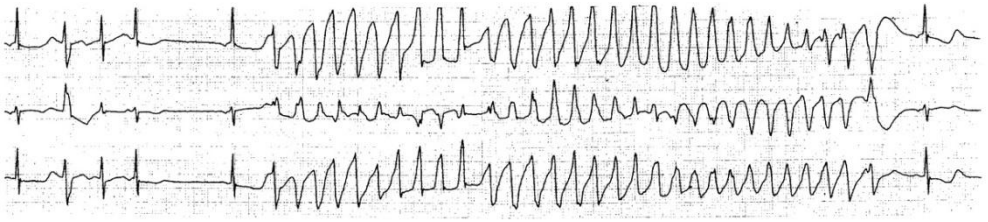
Normal



Pas d'onde P

QRS larges

« Ondulation »



→ Torsade de pointe

Anomalies au niveau du QRS

Normal



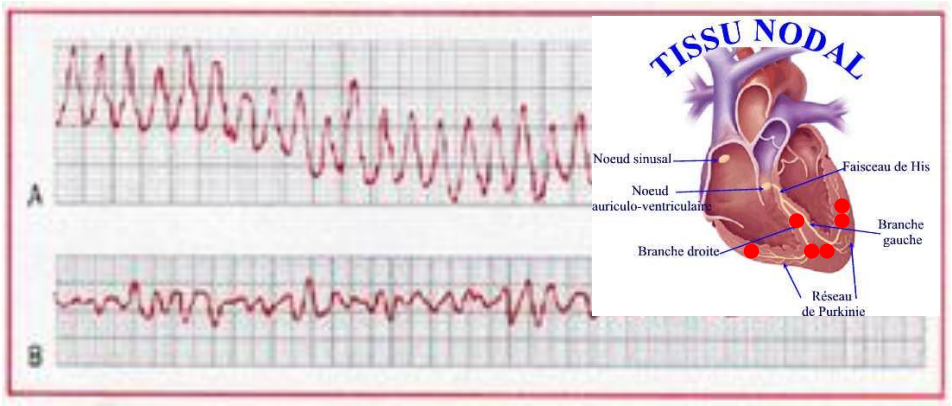


Pas d'onde P

Tracé anarchique

« QRS » large

→ Fibrillation ventriculaire



Anomalies au niveau de l'onde T

Normal





Sus-décalage segment ST

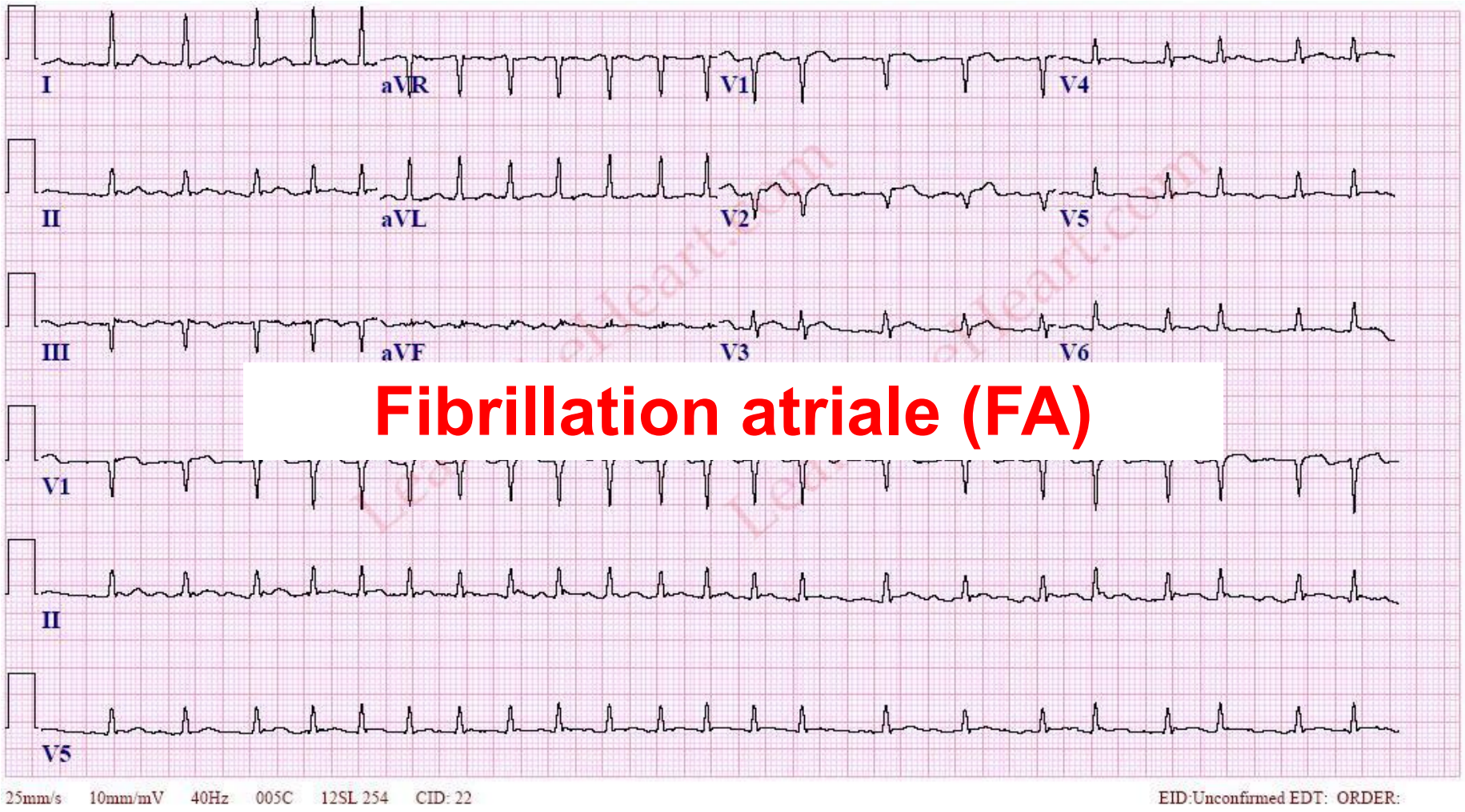
Onde T ample symétrique

Onde Q profonde

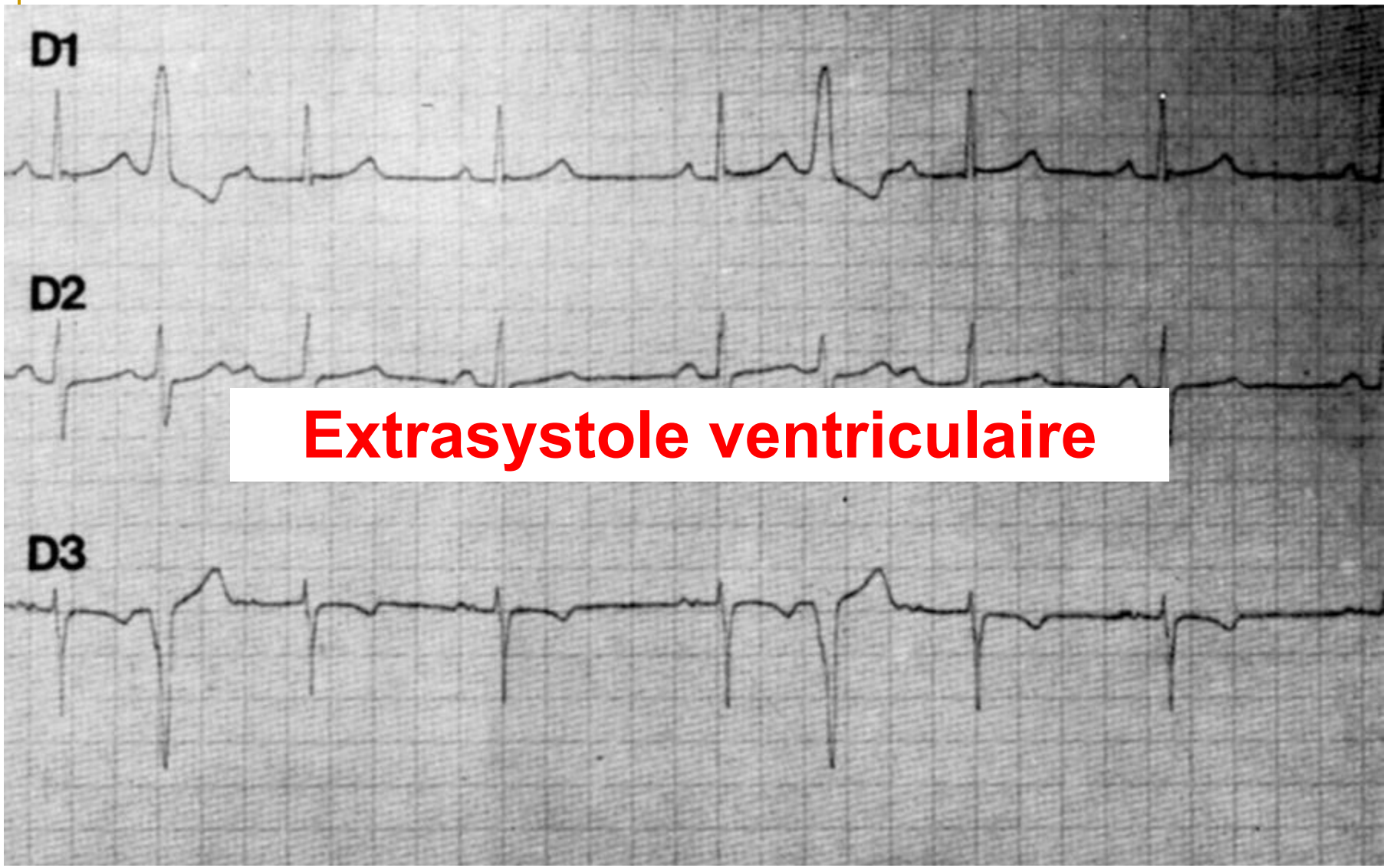
→ Infarctus du myocarde (onde de Pardee)

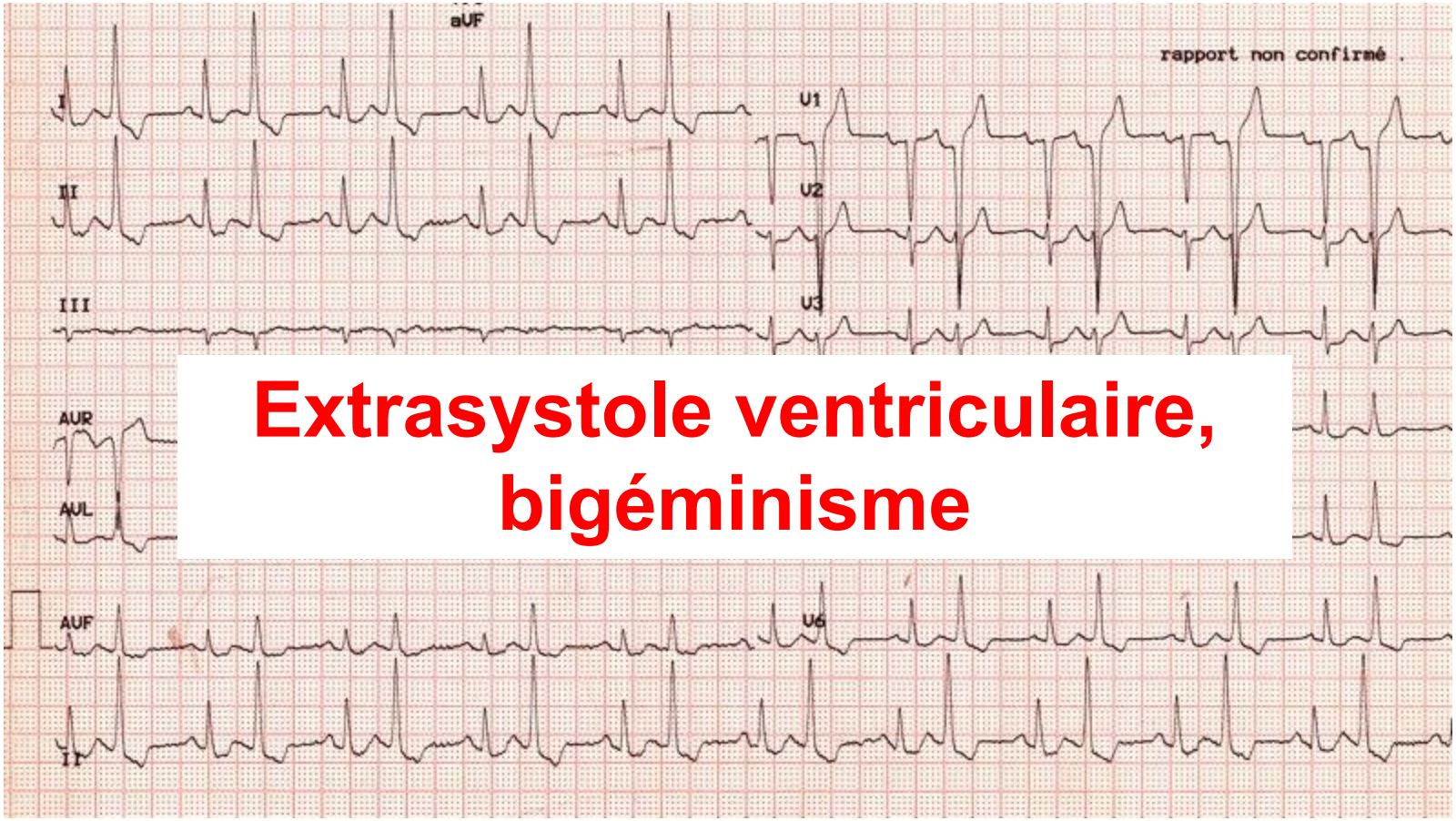


Un peu d’exercice

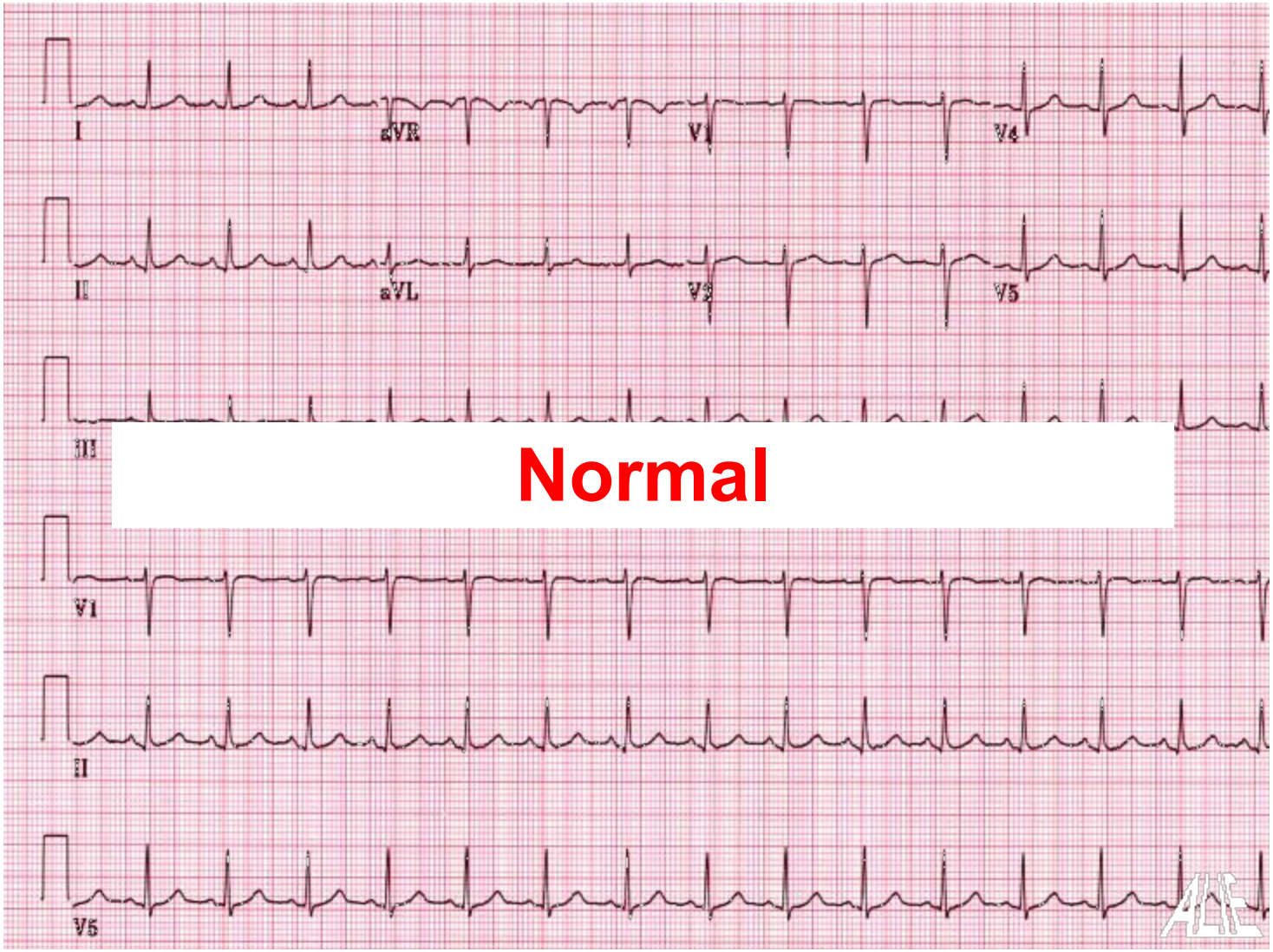


Fibrillation atriale (FA)

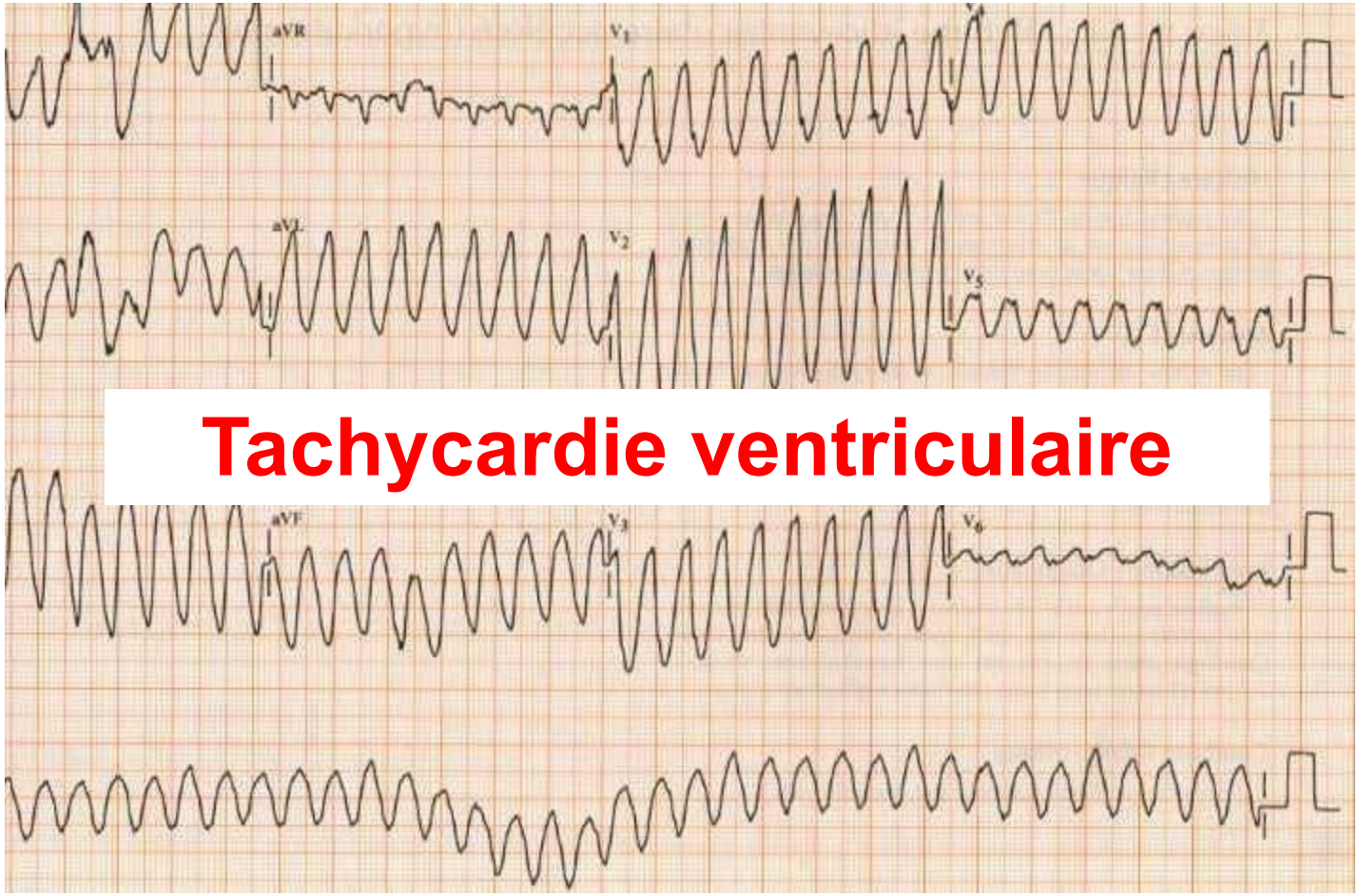

D1
D2
D3
Extrasystole ventriculaire



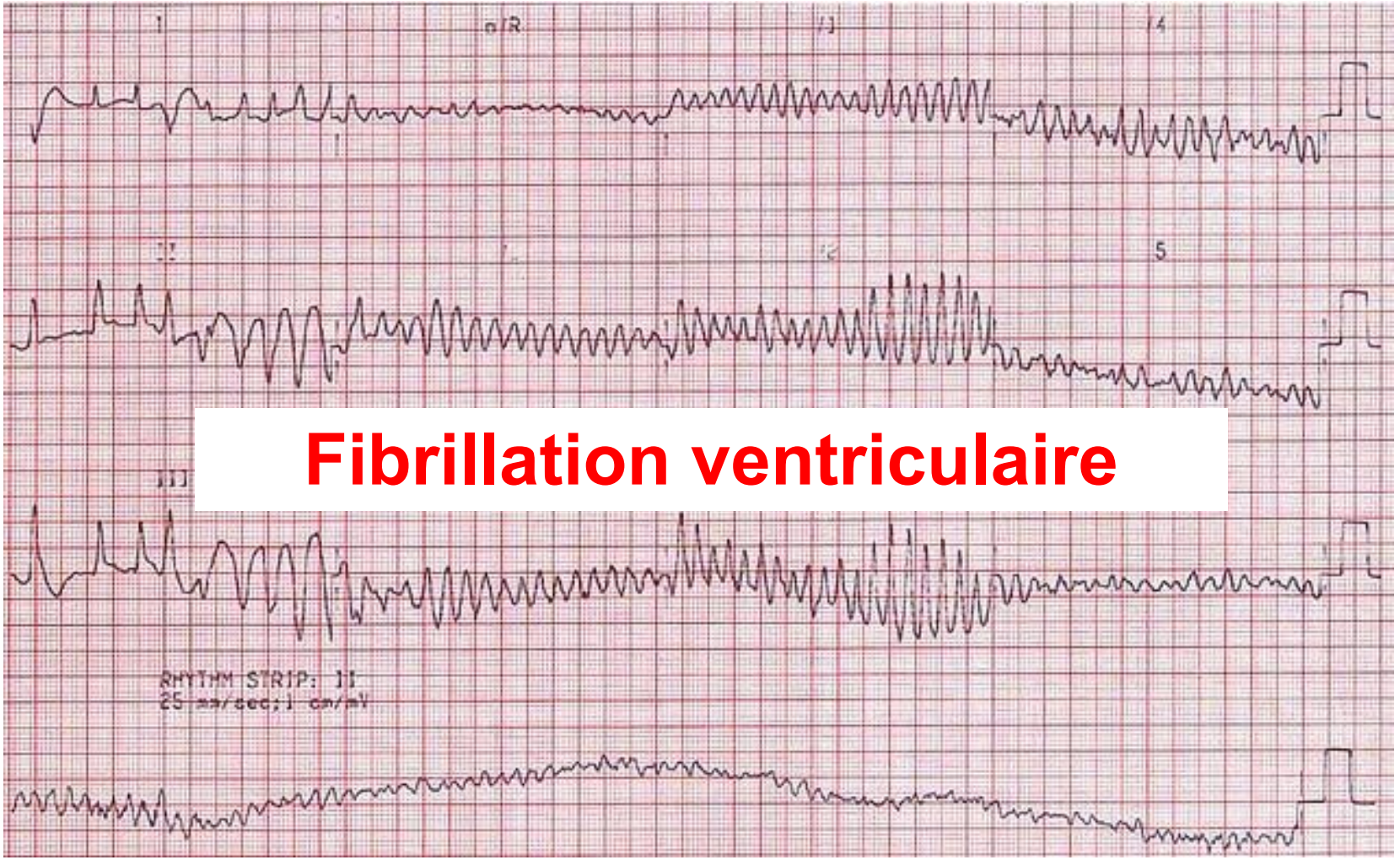
Extrasystole ventriculaire, bigéminisme

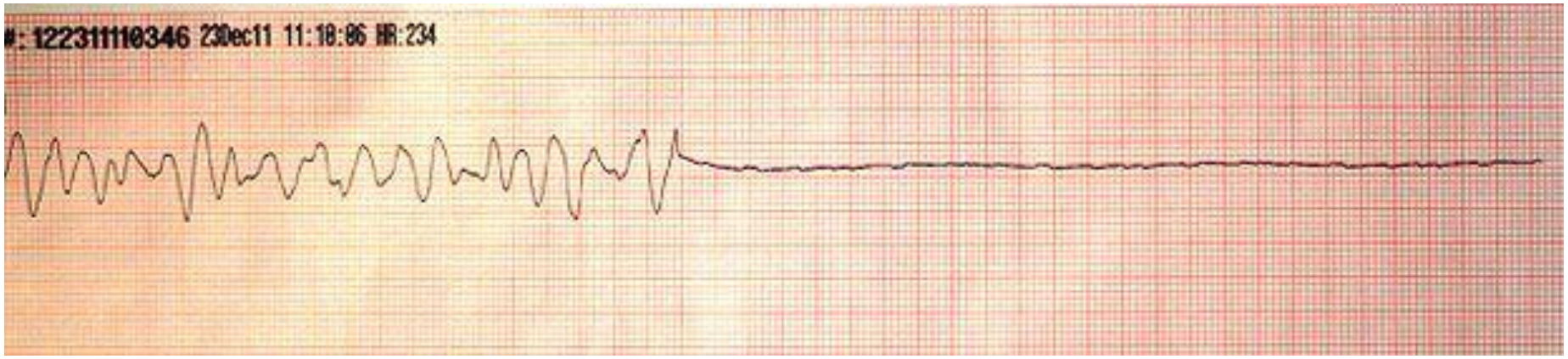


Normal



Tachycardie ventriculaire


Fibrillation ventriculaire
RHYTHM STRIP: II
25 mm/sec; 1 cm/mV



Fibrillation ventriculaire puis asystolie